

Exp: 11 Crypt Arithmetic Problem

Program:

```
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')

for p in permutations(range(10), len(letters)):

    S, E, N, D, M, O, R, Y = p

    if S == 0 or M == 0:

        continue

    SEND = 1000*S + 100*E + 10*N + D

    MORE = 1000*M + 100*O + 10*R + E

    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:

        print("Solution Found")

        print("SEND =", SEND)

        print("MORE =", MORE)

        print("MONEY =", MONEY)

        print("\nLetter Assignments:")

        print(f"S={S}, E={E}, N={N}, D={D}, M={M}, O={O}, R={R}, Y={Y}")

        break
```

Output:

```
Crypt Arithmetic Problem program.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-11 Crypt...
File Edit Format Run Options Window Help
from itertools import permutations
letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
for p in permutations(range(10), len(letters)):
    S, E, N, D, M, O, R, Y = p
    # Leading letters cannot be zero
    if S == 0 or M == 0:
        continue
    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y
    if SEND + MORE == MONEY:
        print("Solution Found")
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        print("\nLetter Assignments:")
        print(f"S={S}, E={E}, N={N}, D={D}, M={M}, O={O}, R={R}, Y={Y}")
        break

Ln: 23 Col: 13
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-11 Crypt Arithmetic Pr
oblem/Crypt Arithmetic Problem program.py
Solution Found
SEND = 9567
MORE = 1085
MONEY = 10652
Letter Assignments:
S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2
>>>

Ln: 12 Col: 0
```